A Point-in-Time-Aware Equity Pairs Research Engine
Statistical Screening, Pair-Specific Tuning, and Ten-Year Evidence

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Abstract

This paper formalizes and evaluates a sector-constrained equity pairs research engine on a frozen snapshot of current S&P 500 constituents. The latest protocol uses ten years of adjusted daily prices, rolling three-calendar-year formation windows, quarterly outer tests, within-sector Benjamini-Hochberg false-discovery-rate control, strict integration and mean-reversion gates, pair-specific cross-validation, next-session execution, terminal liquidation, a 2 basis-point one-way turnover charge, and a 30 basis-point annual short-borrow proxy. Across 28 rebalances, 13,917 different-issuer within-sector pairs were screened at each date. The strict process produced 175 eligible pair-vintages, successfully tuned 174, and evaluated 118 after imposing a maximum of ten per quarter; nine rebalances had no eligible pair. The selected pair-vintages earned a median next-quarter net return of 1.09%, a mean of 1.18%, and a 61.0% positive rate. The frozen baseline's median was 0.47%, and the tuned rule beat it in 57.6% of comparable cases. Cross-sectional training scores contained positive rank information, but the incremental advantage of the chosen rule over the baseline was statistically weak, and a cash-inclusive chronological quarterly series did not establish a profitable portfolio. A separate ten-year identity replay finds attractive ex-post paths for AVY/SW and PPG/SW, but those identities were discovered using the full history and cannot validate a live selection rule. The evidence supports a frozen prospective paper-trading watchlist and continued data collection. It does not support live capital allocation.

Keywords: pairs trading, cointegration, Engle-Granger, false discovery rate, walk-forward validation, transaction costs, selection bias, statistical arbitrage.

1 Introduction

The research question is whether economically similar stocks within the same sector exhibit persistent relative-value relationships that survive chronological selection, signal tuning, turnover costs, and relationship instability. This is harder than finding attractive historical spreads. A broad cross-section creates thousands of candidate pairs, and flexible thresholds create many opportunities to overfit. A valid study must therefore separate three objects:

1. *Statistical diagnosis*: whether a formation-window residual appears stationary and mean reverting.
2. *Executable pair-sleeve evidence*: whether a rule frozen before a quarter earns a subsequent net return.
3. *Deployable portfolio evidence*: whether a contemporaneously selectable collection of sleeves produces robust portfolio-level profit under realistic execution constraints.

The project evolved through several increasingly strict versions. Early work demonstrated that a flattering ex-post winners curve could coexist with a negative broad portfolio. Later work added exact stock-leg accounting, dependence controls, rolling pair-health checks, nested walk-forward selection, and executable-cost scenarios. The current V6 protocol narrows the question further: it tests each statistically eligible pair independently and deliberately does not convert pair-level diagnostics into an aggregate portfolio CAGR.

This paper makes four contributions. First, it translates the implementation into auditable mathematical rules. Second, it documents the complete research evolution, including negative results. Third, it evaluates the predictive content of the training score separately from realized pair returns. Fourth, it distinguishes the latest chronological V6 evidence from a useful but ex-post ten-year identity analysis.

2 Data, Timing, and Evidence Design

2.1 Universe and data lineage

Let U be the frozen set of current S&P 500 constituents and current GICS classifications saved on August 26, 2026. Different share classes belonging to the same issuer are excluded using CIK identifiers. Within each sector g, the candidate set at rebalance date t is

$$\mathcal{P}_{g,t} = \{(i,j) : i,j \in U_g,\ i<j,\ \mathrm{CIK}_i \neq \mathrm{CIK}_j\}. \tag{1}$$

The saved universe generates 13,917 different-issuer within-sector candidates at each V6 screen. Prices are Yahoo Finance adjusted closes. The primary history runs from August 26, 2016 through August 26, 2026, with an end-exclusive boundary of August 27, 2026. The complete ten-year identity replay contains 2,513 sessions.

The snapshot is reproducible but not point-in-time constituent data. Securities that left the index, failed, merged, or changed sector before the snapshot are absent. Consequently, every result is exposed to current-member survivorship and classification bias. Yahoo adjusted closes are suitable for research-level close-to-close tests, not for bid/ask, locate, borrow, or market-impact claims. FRED observations are used only for post-hoc regime attribution; current-vintage macro data never enter pair selection or tuning.

2.2 Rolling clock and information set

At each quarterly rebalance t, the formation set is the exact calendar interval $[t - 3 \text{ years}, t)$. It contains roughly 751–758 observed sessions. The outer test begins at t and ends immediately before the next calendar-quarter boundary; the final window is truncated at the study end. A signal computed using the close at day d changes holdings on day $d + 1$.

All outer-state variables reset to flat, and each outer test is forcibly liquidated at its terminal close. This prevents hidden carry between independent pair-vintages. Screening, orientation, hedge ratio, hyperparameters, and cost assumptions are frozen before the outer return is observed.

Table 1: Evidence layers and their permissible interpretation.

Layer	Unit	Design	Interpretation
Initial 8-year study	Portfolio and pair	5-year formation, 3-year held-out test, 110 slots	Baseline diagnostic; quota fills later invalidated
Production-control study	Ten-pair portfolio	Exact stock legs, rolling allocator, health and execution gates	Research/paper trade only; several negative gates
V6 primary study	Pair-vintage	Rolling 3-year formation, pair-specific tuning, next-quarter test	Main chronological evidence; no portfolio CAGR
Ten-year identity replay	Fixed pair identity	Full-calendar stitched quarterly replays	Ex-post descriptive ranking only

3 Statistical Screening and Pair Construction

3.1 Two-orientation Engle-Granger screen

For adjusted prices $P_{i,d}$ and $P_{j,d}$, the engine fits both log-price orientations on the formation window. For the chosen orientation,

$$\log P_{i,d} = \alpha_{ij,t} + \beta_{ij,t} \log P_{j,d} + \varepsilon_{ij,d}, \tag{2}$$

where $(\alpha_{ij,t}, \beta_{ij,t})$ are estimated only from formation observations. The pair-level orientation-adjusted p-value is

$$p_{ij,t}^{\text{pair}} = \min\left(1, 2\min\{p_{i|j,t}, p_{j|i,t}\}\right). \tag{3}$$

The factor of two corrects for choosing the better of two orientations. Within each sector, the pair-level values are converted to Benjamini-Hochberg q-values at a 5% false-discovery-rate threshold [5]. Both component log-price series must look $I(1)$, the selected hedge ratio must be positive, residual half-life must be between 2 and 252 sessions, the Hurst exponent must be below 0.50, and at least 90% of formation sessions must be observed. No quota fill or exploratory fallback is permitted.

This procedure builds on the Engle-Granger representation theorem [1] and the economic intuition behind convergence trading [2]. Cointegration is treated as a conditional diagnostic. It is not assumed to be structurally permanent.

3.2 Residual signal and position state

Given a frozen formation hedge relation, the outer residual is

$$s_{ij,d} = \log P_{i,d} - \widehat{\alpha}_{ij,t} - \widehat{\beta}_{ij,t} \log P_{j,d}. \tag{4}$$

For an SMA rule with lookback L,

$$z_{ij,d} = \frac{s_{ij,d} - \overline{s}^{(L)}_{ij,d}}{\widehat{\sigma}^{(L)}_{ij,d}}, \tag{5}$$

with an analogous exponentially weighted estimator for EWMA configurations. The state machine enters short residual when z exceeds the positive entry threshold and long residual when z falls below its negative threshold. It exits when $|z|$ returns below the exit threshold, when $|z| \geq 4$, or when the maximum holding period is reached.

The pair sleeve uses one-times gross exposure. Before normalization, its canonical leg vector is $(1, -\widehat{\beta})$. With sign $q_{ij,d} \in \{-1, 0, 1\}$,

$$(w_{i,d}, w_{j,d}) = q_{ij,d} \frac{(1, -\widehat{\beta}_{ij,t})}{1 + |\widehat{\beta}_{ij,t}|}, \qquad |w_{i,d}| + |w_{j,d}| \leq 1. \tag{6}$$

This hedge is a cointegration coefficient. It does not guarantee market-, sector-, or factor-neutrality.

3.3 Trading costs

Let $r_{k,d+1}$ be the next close-to-close return for leg k, and let $\widetilde{w}_{k,d}$ be its post-return pretrade weight. One-way leg turnover is

$$\mathrm{TO}_{d+1} = \frac{1}{2} \sum_{k \in \{i,j\}} |w_{k,d+1} - \widetilde{w}_{k,d}|. \tag{7}$$

The pair-sleeve net return is

$$R^{\mathrm{net}}_{d+1} = \sum_{k} w_{k,d} r_{k,d+1} - c\,\mathrm{TO}_{d+1} - \sum_{k: w_{k,d} < 0} b_d |w_{k,d}|, \tag{8}$$

where $c = 2\,\mathrm{bp}$ per dollar traded in the primary test and the annual short-borrow proxy is 30 bp. Launch, hedge drift, signal changes, and terminal liquidation are charged. A frozen 1 bp sensitivity reprices the selected rules without retuning.

3.4 Pair-specific tuning

Each strict pair searches 96 predeclared configurations plus one unchanged baseline. The grid crosses SMA/EWMA, lookbacks of 10/20/40/60 sessions, entry levels of 1.25/1.75/2.25, exit levels of 0.25/0.50, and maximum holds of 10/30 sessions. Inner validation uses up to five expanding folds with 378 training sessions, 63 validation sessions, and a one-session purge. Each fold starts and ends flat; hedge relations are refit using training observations only.

The optimization objective is pair-level net CAGR. A one-standard-error rule prefers lower fold dispersion, then lower turnover, shallower drawdown, higher pooled validation CAGR, and deterministic configuration ID. A selected configuration must have at least 10 full-training trades, 6 pooled validation trades, and 3 active folds. The quarter then ranks successful pairs by cross-validated net CAGR and keeps at most ten; fewer than ten and zero are valid outcomes.

4 Performance Measurement

For daily net returns R_d and D sessions, wealth is $W_D = \prod_{d=1}^{D}(1 + R_d)$. The reported metrics are

$$\mathrm{CAGR} = W_D^{252/D} - 1, \tag{9}$$

$$\sigma_{\mathrm{ann}} = \sqrt{252}\,\mathrm{sd}(R_d), \tag{10}$$

$$\mathrm{Sharpe} = \sqrt{252}\,\frac{\overline{R}}{\mathrm{sd}(R_d)}, \tag{11}$$

$$\mathrm{MaxDD} = \min_d \left(\frac{W_d}{\max_{u \leq d} W_u} - 1 \right). \tag{12}$$

The primary V6 outcome is the actual next-quarter net total return of a pair-vintage. Cross-sectional predictive content is measured with quarterly Spearman rank information coefficients. Because quarterly cross-sections vary in size and formation windows overlap, the analysis also reports Newey-West statistics, within-quarter permutations, and four-quarter block-bootstrap intervals [6]. These are diagnostics, not substitutes for an independent prospective sample.

5 Research Evolution Before V6

The initial research package deliberately separated descriptive winners from causal tests. The equal-capital portfolio of 110 formation-selected sleeves earned -0.46% net CAGR with a -4.62% maximum drawdown. The ten best pairs selected after seeing the held-out test earned +12.40% CAGR, but a circular-block null experiment showed that this result was ordinary after repeating the winner-selection step. Only 7 of the 110 original slots had passed every strict formation gate; 103 were quota fills.

Follow-up work then tested allocation and production controls. On a common final-year exact stock-leg ledger, concentrating into ten forward-selected pairs hurt performance: equal capital earned -2.80% CAGR, while constrained equal-risk contribution improved the outcome to -1.61%. A rolling expected-return floor worsened timing. A production-style health layer ended with 0 of 10 pairs eligible, while a five-fold purged nested replay earned -3.57% CAGR. The later strict-only correction produced +0.30% CAGR with a wide bootstrap interval. These results motivated the V6 decision to stop quota filling, isolate pair-vintage evidence, and avoid a portfolio claim.

Table 2: Selected research milestones. Results use different designs and are not directly additive.

Experiment	Reported result	Main lesson	Status
All 110 held-out sleeves	-0.46% CAGR	Broad forced selection lacked edge	Diagnostic
Ex-post best ten	+12.40% CAGR	Winner selection can manufacture a strong curve	Selection-biased
Final-year selected ten, static ERC	-1.61% CAGR	Unequal risk helped, signal remained unprofitable	Retrospective
Production nested OOS	-3.57% CAGR	Exposure controls did not create return	Do not deploy
Corrected strict-only nested replay	+0.30% CAGR	Small positive estimate with wide uncertainty	Paper trade only
V6 pair-vintage protocol	1.09% median quarter	Positive local evidence, no aggregate portfolio claim	Primary current study

6 V6 Chronological Pair-Vintage Results

6.1 Strict candidate supply

Figure 1 shows that statistical opportunity was highly regime dependent. Across 28 rebalances, 175 strict pair-vintages appeared, but nine quarters had none. Candidate supply peaked at 28 strict pairs in January 2022, while several 2024–2026 dates supplied zero. The maximum-ten rule therefore binds only in dense periods; zero exposure is a normal outcome of the strategy design.

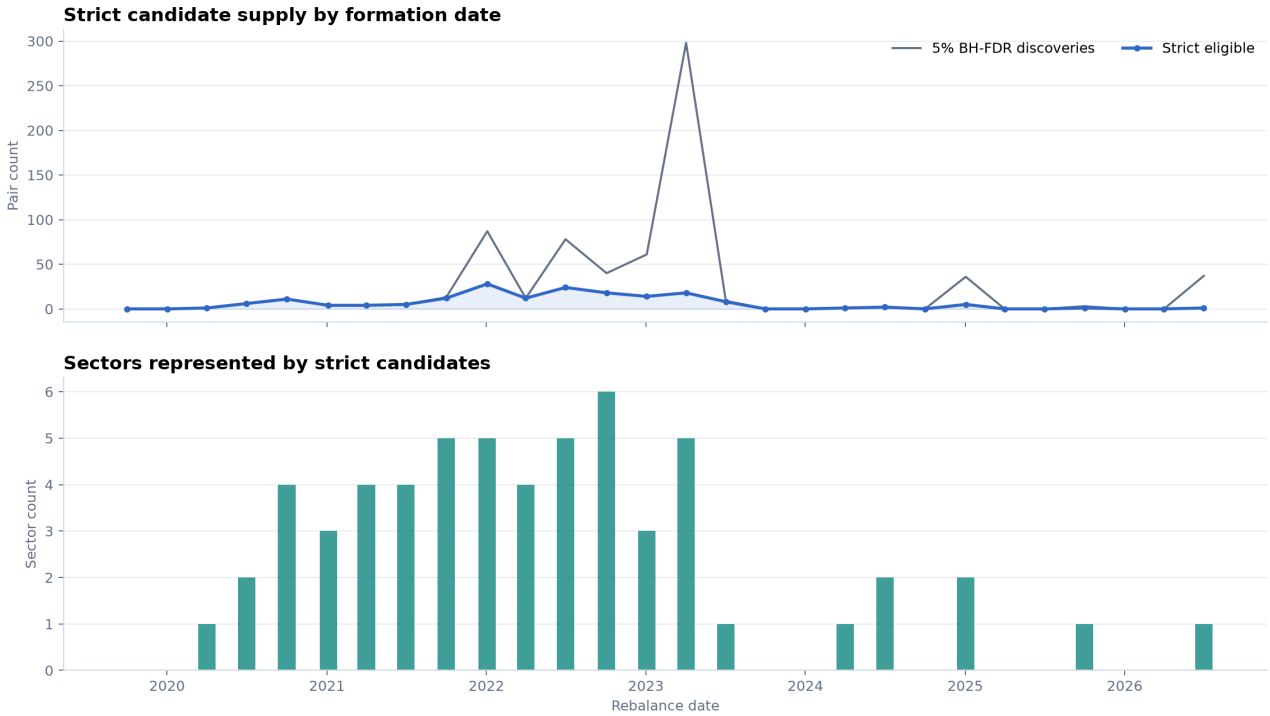


Figure 1: Strict candidate supply by formation date. The gray line is the count passing the 5% within-sector BH-FDR gate; blue adds all strict integration, hedge-ratio, half-life, Hurst, and coverage gates. Source: saved V6 artifacts.

6.2 Headline scorecard

Of 175 strict candidates, 174 completed tuning and 118 were selected under the quarterly top-ten cap. The primary 2 bp results are summarized in Table 3. The average selected pair-vintage earned 1.18% during its next quarter, the median earned 1.09%, and 61.0% were positive. The median annualized pair-sleeve CAGR was 4.45%, but quarter-level total return is the more stable interpretation because each outer window contains only about one quarter.

The tuned configuration outperformed the unchanged baseline in 57.6% of comparable cases. The median tuned-minus-baseline total-return difference was +0.55 percentage points. The 1 bp frozen sensitivity raised the median pair-vintage return from 1.09% to 1.18%, leaving the positive fraction unchanged. Trading was sparse: the median sleeve recorded three entries, while median annualized leg turnover was about 25.4 times capital.

Table 3: V6 outer-test pair-vintage scorecard.

Metric	Tuned, 2 bp	Baseline, 2 bp	Tuned, frozen 1 bp
Pair-vintages	118	118	118
Mean next-quarter net return	1.18%	0.36%	1.25%
Median next-quarter net return	1.09%	0.47%	1.18%
Positive fraction	61.0%	55.1%	61.0%
Median annualized pair CAGR	4.45%	1.90%	–
Tuned rule beats baseline	57.6%	–	–
Median entries per pair-vintage	3	–	3

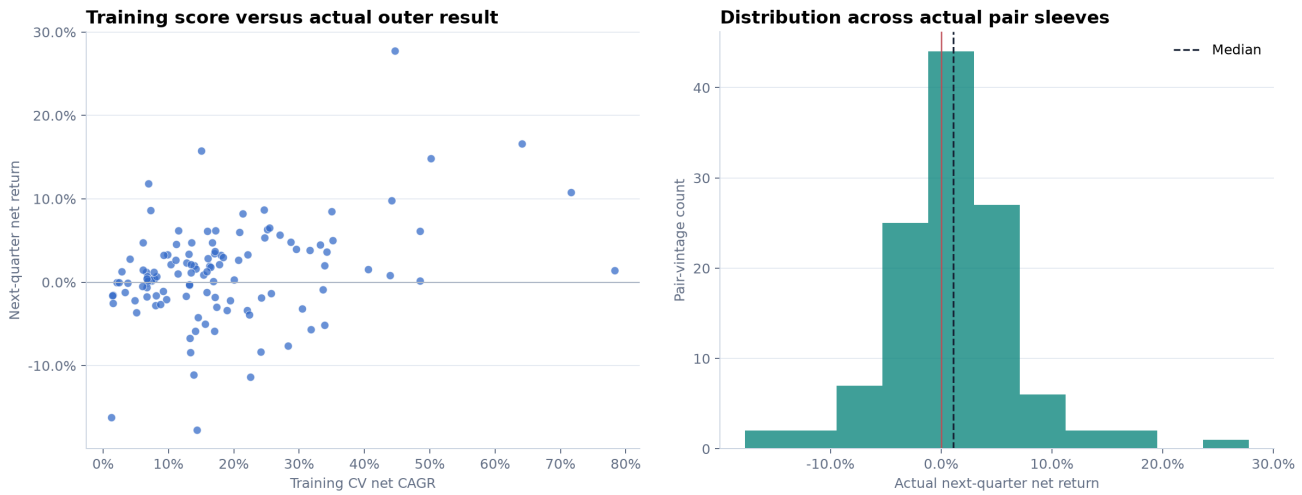


Figure 2: Training cross-validation score versus actual next-quarter return, and the distribution of actual selected pair-vintage returns. The relationship is positive but noisy, with economically meaningful negative outliers. Source: saved V6 artifacts.

6.3 Predictive content and portfolio-level caution

The training score has measurable cross-sectional information. Among quarters with at least five eligible pairs, the mean quarterly Spearman IC is 0.250, with a four-quarter block-bootstrap 95% interval of [0.150, 0.354]. Across all tuned strict candidates, the pooled Spearman correlation is 0.277. This supports the claim that stronger training scores tended to rank stronger next-quarter pair returns in the observed sample.

The optimizer's incremental contribution is less secure. When the target is the selected rule's advantage over the unchanged baseline, the mean quarterly IC is 0.034 for quarters with at least five pairs, with a block interval of [-0.172, 0.222]. The tuned rule therefore improved average outcomes, but its relative training score did not reliably rank where tuning would help most.

Pair-vintage averages also do not equal a deployable portfolio. The selected top-ten mean return across the 19 active quarters was -0.11%, with a block interval of [-2.75%, 1.71%]. When nine zero-supply quarters are retained as cash, the mean across all 28 scheduled quarters was -0.08%, with a block interval of [-1.75%, 1.30%]. These are equal-quarter diagnostics rather than a fully specified capital-weighted portfolio, but they show why the positive pair-vintage median must not be promoted into a portfolio CAGR.

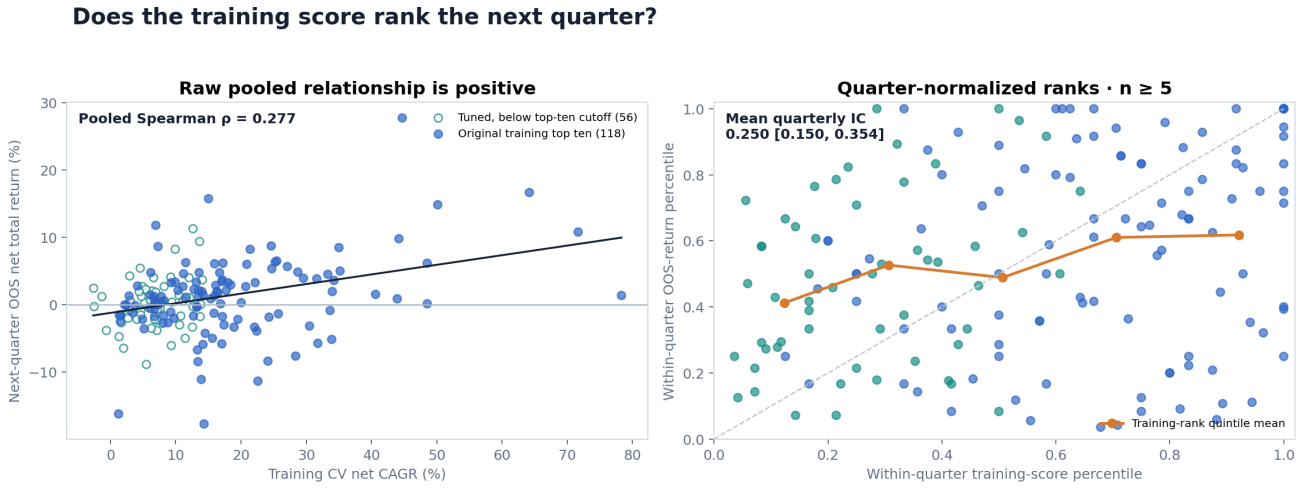


Figure 3: Predictive diagnostic for the pair-level training score. The right panel normalizes ranks within quarter, which is the relevant cross-sectional comparison. The analysis is post-hoc and based on a short quarterly sample. Source: saved predictive-diagnostics artifacts.

7 Ten-Year Identity Replay: Descriptive Leaders

The identity study fixes all 114 unique pair identities discovered anywhere in the strict-and-tuned history, replays each identity across the same 41 quarterly intervals, and keeps unavailable, failed, or zero-trade quarters as cash. This creates a common complete-calendar denominator. It is useful for persistence and downside analysis, but the identity universe itself was learned from the full sample. Every ranking in this section is ex-post.

At the primary 2 bp cost, 80 of 114 identities have positive ten-year CAGR, 23 have a positive year-cluster bootstrap lower bound, and 56 remain positive after both their best quarter and best complete calendar year are separately set to cash. The strongest full-calendar path is AVY/SW: 21.52% CAGR, 1.31 Sharpe, -18.00% maximum drawdown, and a 95% year-cluster CAGR interval of [13.02%, 30.38%]. PPG/SW has the strongest downside profile among the leaders: 13.44% CAGR, 0.99 Sharpe, -12.57% maximum drawdown, and an interval of [8.33%, 18.49%].

Table 4: Top complete-calendar pair identities at 2 bp. All rows are ex-post descriptive.

Pair	CAGR	Sharpe	Max DD	95% CAGR low	Ex-best qtr	Ex-best year
AVY / SW	21.52%	1.31	-18.00%	13.02%	19.49%	16.69%
PPG / SW	13.44%	0.99	-12.57%	8.33%	11.79%	9.78%
AMZN / CVNA	13.43%	0.91	-22.42%	2.78%	10.68%	7.38%
BLDR / FERG	12.03%	0.78	-24.13%	2.68%	10.00%	8.86%
CRH / SW	11.79%	0.78	-37.59%	2.32%	9.02%	7.56%
AME / FERG	8.48%	0.75	-12.60%	5.06%	7.06%	6.47%

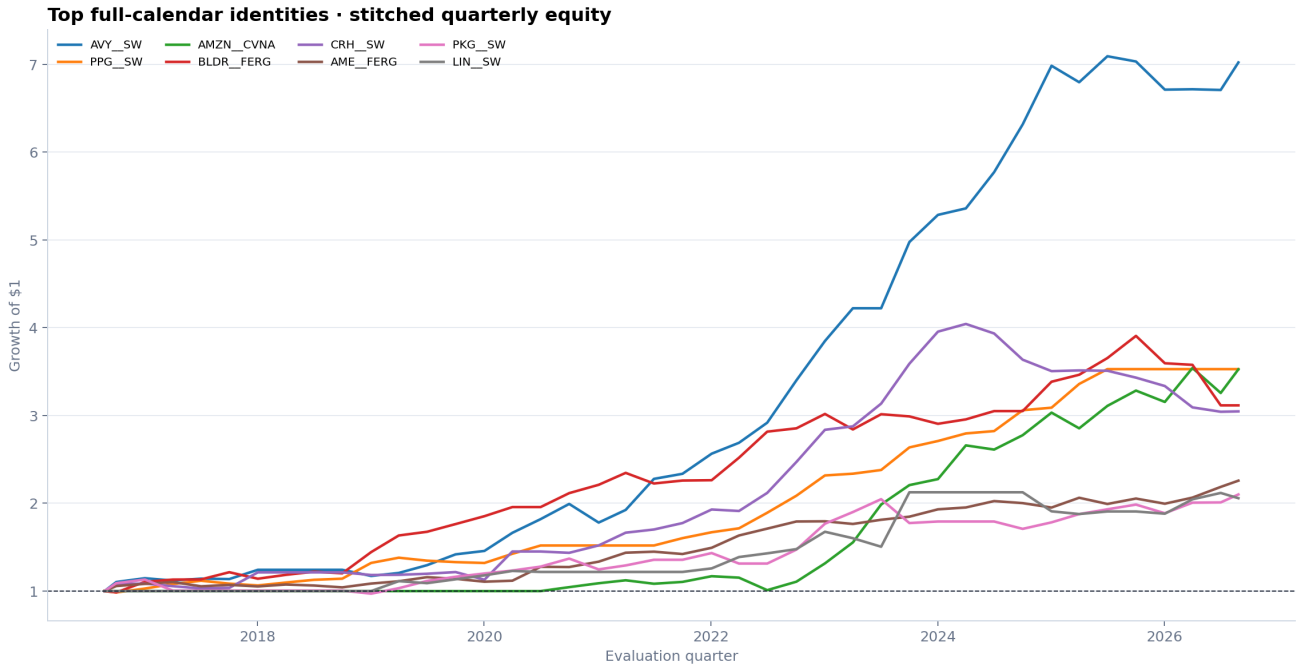
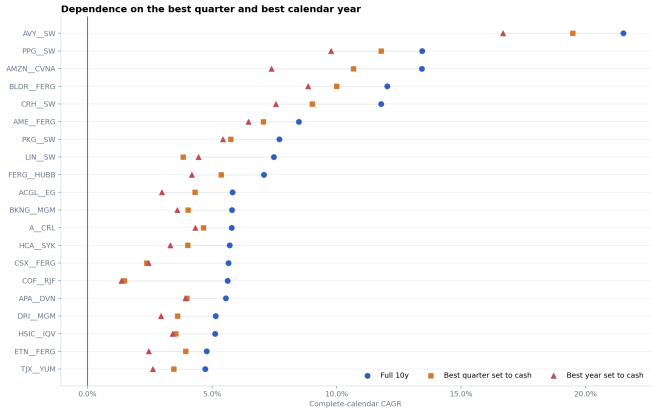
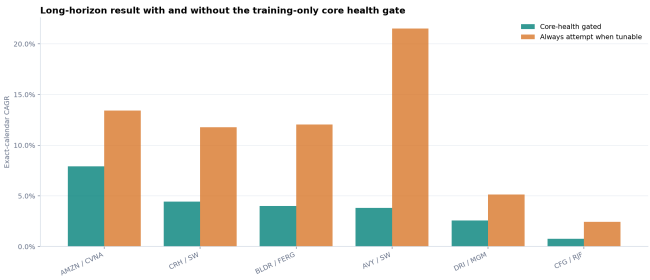


Figure 4: Stitched complete-calendar equity curves for leading identities. Cash quarters remain in the denominator. Identity membership is ex-post and these curves are not a historical portfolio selection rule. Source: saved identity-evaluation artifacts.

The strongest curves do not imply stable statistical eligibility. BLDR/FERG has the highest strict-eligibility count among the leaders, yet it passes in only 7 of 41 intervals. AVY/SW passes in 3, PPG/SW in 1, and the maximum for any identity is 7. A separate training-only core-health gate sharply reduces calendar CAGR for the six originally highlighted identities because it forces cash in many windows. This is economically sensible: a pair can earn attractive stitched returns when tunable while failing the stricter definition of persistent relationship health.



(a) Best-period removal.



(b) Training-only core-health gate.

Figure 5: Robustness diagnostics for ex-post leaders. Left: setting the best quarter or year to cash reduces but does not eliminate the strongest paths. Right: enforcing the core-health gate materially reduces activity and return. Source: saved identity and leader-replay artifacts.

8 Economic Interpretation

8.1 What appears to work

The strongest current evidence is local and cross-sectional. Within quarters that supplied several strict pairs, the training score tended to rank next-quarter returns, and the top-ten cap outperformed nonselected strict pairs in eight comparable quarters. Several recurring identities, including CRH/SW, AME/FERG, BLDR/FERG, and AVY/SW, posted multiple positive vintages. The result is consistent with episodic convergence opportunities that become more valuable when the formation relationship is unusually strong.

The one-standard-error rule also appears useful as a regularizer. It avoids selecting the numerically best but unstable configuration, and the cost sensitivity is modest between 1 and 2 bp. These properties support continued prospective observation.

8.2 What has not been established

The evidence does not establish a scalable, continuously invested portfolio. Strict candidate supply is intermittent, many apparently strong identities are rarely eligible, and mean returns across the full scheduled-quarter clock are close to zero. The incremental predictive content of choosing among 97 signal configurations is weak. Earlier exact-ledger portfolio experiments were negative after concentration, dynamic allocation, and production gates.

The gap between pair-sleeve success and portfolio success has several mechanisms. Pairs can share underlying names and sectors, so sleeve returns are dependent. Candidate abundance clusters in regimes, causing risk to rise when many pairs qualify. Hedge ratios neutralize a log-price relation, not market beta. High annualized leg turnover can turn a modest spread forecast into a capacity-sensitive execution problem. Finally, the same current-member universe is reused across the historical period.

9 Robustness, Limitations, and Required Validation

The saved implementation passes chronology, eligibility, cost-freeze, and artifact-reconciliation checks. Those checks validate the engine's invariants, not the profitability hypothesis. The following limitations remain material:

- **Point-in-time universe.** Replace current constituents and GICS labels with dated membership, classifications, delisted securities, and corporate-action histories.
- **Execution data.** Replace adjusted-close fills with historical NBBO or trade data, observed spreads, ADV, market impact, borrow fees, and locate availability. A 2 bp charge is a sensitivity, not a certified fill model.
- **Independent sample.** Freeze the watchlist, screening thresholds, tuning grid, and ranking rule before collecting new quarters. Do not add identities after observing their returns.

- **Multiplicity across research versions.** The project explored screening, signal, allocation, and health variants. Future inference should account for the entire research path, not only the final 97-cell within-pair grid [7].
- **Dependence.** Use quarter or rebalance date as the independent unit. Cluster or block resampling must preserve shared names, sectors, and overlapping formation histories.
- **Portfolio formation.** Predeclare how scarce eligible pairs share capital, how shared underlyings are capped, how cash is treated, and how beta/factor exposure is constrained. Evaluate the full rule chronologically.
- **Regime and structural breaks.** Calibrate health thresholds using training-only data, then test whether the gate improves future return rather than merely reducing exposure.
- **Benchmarking.** Compare a frozen portfolio to cash, a market-neutral factor benchmark, and simple distance-based pairs [2]. Report alpha, beta, turnover, gross exposure, drawdown, and capacity together.

The minimum production gate is therefore stricter than a positive pair-vintage median. A candidate strategy must produce a positive portfolio-level net return with uncertainty bounded away from zero, stable performance across rebalance dates and sectors, acceptable drawdown, no material capacity breaches at target capital, and fully point-in-time inputs.

10 Conclusion

The current research engine is substantially more credible than the original forced-quota study. It screens every different-issuer within-sector pair, controls within-sector false discoveries, requires integration and mean-reversion diagnostics, tunes each pair using only formation data, executes one session later, charges turnover and borrow, and preserves zero-supply quarters. The latest V6 evidence is encouraging at the pair-vintage level: 118 selected outer tests produced a 1.09% median next-quarter return, a 61.0% positive rate, and a positive cross-sectional relationship between training score and future return.

The deployment conclusion remains conservative. Pair-level evidence did not translate into a demonstrated chronological portfolio; the optimizer's incremental ranking advantage was weak; earlier exact-ledger portfolio experiments were negative; and the attractive ten-year identity leaders were selected ex-post. AVY/SW and PPG/SW are appropriate anchors for a frozen prospective watchlist, with other recurring identities retained as diagnostics. No identity should receive live capital based on the present report.

The practical next step is a prospective paper-trading phase using point-in-time membership, frozen rules, execution-grade observations, and a predeclared portfolio allocator. New data should be treated as the test set, not as another tuning window.

A Reproducible Walk-Forward Checklist

At every quarterly rebalance t, a valid run should execute the following sequence:

1. Freeze the dated universe, GICS labels, issuer identities, price snapshot, and configuration hashes.
2. Construct the exact three-calendar-year formation window ending before t.
3. Test $I(1)$ behavior and both Engle-Granger orientations using formation data only.
4. Apply the orientation correction, within-sector BH-FDR, coverage, positive-beta, half-life, and Hurst gates.
5. Tune every strict pair over the frozen 97-configuration set; fail closed when trade or fold requirements are not met.
6. Rank successful pairs using formation-only cross-validated net CAGR; keep at most ten and permit zero.
7. Freeze hedge ratios, signal parameters, and costs before opening the outer quarter.
8. Apply close-d signals on $d+1$, reconstruct both stock legs, and charge launch, drift, signal, borrow, and liquidation costs.
9. Save daily positions, signals, trades, turnover, costs, failures, and hashes. Reconcile every summary to the ledger.
10. Keep pair-vintage, portfolio, and ex-post identity conclusions in separate output namespaces.

B Data and Artifact Availability

The paper was assembled from saved local artifacts. The primary inputs are the V6 pair-vintage metrics, baseline metrics, cost sensitivity, predictive-diagnostics summaries, ten-year identity metrics, complete-calendar daily returns, trade ledgers, quality

checks, and manifests. The principal human-readable reports are:

- pair_optimized_10y_3y_report.html
- pair_predictive_diagnostics_report.html
- pair_identity_10y_evaluation_report.html
- research_evolution_report.html
- fallback_credibility_report.html
- production_gates_report.html

The corresponding JSON manifests contain SHA-256 hashes and explicitly record that credentials were not serialized. Quality-check status is scoped to implementation, chronology, eligibility, and reconciliation. It is not a statistical deployment pass.

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